CLAIMS

What is claimed is:

1. A system for countering covert surveillance in a physical environment, the system comprising:

   (a) A plurality of distributed sensor nodes, each node configured to:

      i. Detect short-range radiofrequency (RF) signals indicative of interrogation of covert listening devices.

      ii. Detect visual anomalies indicative of covert camera surveillance by mobile devices.

      iii. Monitor network traffic patterns for indications of covert mesh network communications.

   (b) A central analysis and control unit (CACU) communicatively coupled to the plurality of distributed sensor nodes, the CACU configured to:

      i. Aggregate detection data from the sensor nodes.

      ii. Execute machine learning algorithms to identify patterns of covert surveillance activities based on the aggregated data.

      iii. Assess a threat level associated with identified covert surveillance activities.

   (c) A neutralization module communicatively coupled to the CACU, the neutralization module configured to perform one or more countermeasure actions in response to an assessed threat, wherein said actions include at least one of localized RF jamming, data obfuscation, or alert generation for direct intervention.

2. The system of claim 1, wherein the short-range RF signals include Near Field Communication (NFC) signals.

3. The system of claim 1, wherein the mobile devices are smartphones with deep hardware integrations enabling interrogation of the covert listening devices.

4. The system of claim 1, wherein the visual anomalies indicative of covert camera surveillance include a mobile device camera pointed at a person's face or a shopping basket.

5. The system of claim 1, further comprising a regulatory framework for the deployment and use of the system, said framework including a provision for a perpetual, non-exclusive, irrevocable, royalty-free, worldwide license granted to all national, state, and local governmental entities for the use, manufacture, and distribution of the countermeasure technology.

6. A method for countering covert surveillance in a physical environment, the method comprising the steps of:

   (a) Deploying a plurality of distributed sensor nodes within the physical environment.

   (b) Utilizing the sensor nodes to detect:

      i. Short-range radiofrequency (RF) signals indicative of interrogation of covert listening devices.

      ii. Visual anomalies indicative of covert camera surveillance by mobile devices.

      iii. Network traffic patterns for indications of covert mesh network communications.

   (c) Aggregating detected data at a central analysis and control unit (CACU).

   (d) Employing machine learning algorithms at the CACU to identify patterns of covert surveillance activities and assess an associated threat level.

   (e) In response to an assessed threat, performing one or more countermeasure actions using a neutralization module, wherein said actions include at least one of localized RF jamming, data obfuscation, or alert generation for direct intervention.

7. The method of claim 6, further comprising operating under a regulatory framework that includes granting a perpetual, non-exclusive, irrevocable, royalty-free, worldwide license to all national, state, and local governmental entities for the use, manufacture, and distribution of the countermeasure technology.

8. The method of claim 6, wherein the covert listening devices are embedded in consumer products or infrastructure.

9. The method of claim 6, wherein the network traffic monitoring identifies communications that bypass traditional network protocols.

10. The method of claim 6, wherein the countermeasure action of data obfuscation includes injecting decoy data into detected covert communication channels.